Supplementary Information

A nationwide integrated demand–supply framework for analyzing urban carbon footprints and mitigation potential in Israel

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Supplementary Notes

Supplementary Note 1. Extended details for electricity footprint calculations

The electricity footprint quantifies the GHG emissions associated with urban electricity consumption, covering both residential and public-commercial sectors. Electricity consumption data (kWh) of national-grid electricity, the primary source of electricity consumed across the country, was obtained from the Israeli Electric Company (IEC, 2023). Consumption data of electricity provided by Independent Power Producers (IPPs) was received from the Israeli Ministry of Energy's report (MOE, 2023)

Next, the data was refined to exclude water and wastewater infrastructure consumption, as these are already accounted for in their own respective footprint categories. Industrial electricity consumption was also excluded due to its export nature, in line with CBCF principles. Ultimately, the final dataset covered households, commercial, and public sectors. For the analysis, the commercial and public sectors were merged into one sector, as it was often found that consumption data classified as the public sector was actually the commercial sector consumption or vice versa.

Initial emissions were calculated by multiplying electricity consumption (kWh) by emission factors (EFs) for both direct power generation and upstream processes, tailored to the specific electricity providers serving each city. Notably, the emission intensity of the national grid is higher than that of IPPs due to differences in generation sources. This approach ensures that computed emissions accurately reflect the actual mix of sources supplying each city.

Local photovoltaic (PV) generation was incorporated into the analysis, assuming substitution of electricity otherwise supplied by the national grid and IPPs. Based on installed capacity values, local PV production and the corresponding avoided GHG emissions were estimated (Martire et al., 2018).

The accounting steps were as follows:

1. **Calculating PV electricity production**: Data on local PV installed capacity for all analyzed cities was obtained from the National Electric Authority (TEA, 2023) in kilowatt direct current (kW DC). This was converted to kilowatt-hour alternating current (kWh AC) by multiplying by 1668 (TEA, 2020) and then by 0.98 to account for DC-to-AC conversion efficiency.

2. **Determining the PV share of total consumption (ratio)**: For each city, the ratio of PV production (kWh) to the total electricity consumption (kWh) was calculated. This ratio represents the share of the city's electricity demand met by PV generation.

3. **Calculating the emissions reduction factor**: The emissions reduction factor was defined as 1 minus the PV share of total consumption (the ratio found earlier). This factor reflects the proportion of emissions that are not avoided due to PV use (In other words, it represents the proportion of emissions not offset by PV).

4. **Applying the reduction factor**: The factor was applied to the initial emissions estimates, yielding net emissions after accounting for avoided GHGs from PV generation.

Emission Factors Calculation Procedure

Direct emission factors

Direct emission factors (tCO_2eq/kWh) for each electricity provider were calculated by dividing the total emissions from each power generation station (MOEP, 2023) by the total electricity produced by that station (TEA, 2021).

Upstream process emission factors

Upstream process EFs account for emissions from natural gas extraction, natural gas transportation, and coal extraction. The main steps for each component are summarized below:

Natural Gas Extraction:

- Methane emissions from gas fields were obtained from the national Pollutant Release and Transfer Register (MOEP, 2023).
- The quantity of natural gas used for electricity production was obtained from the Central Bureau of Statistics (CBS, 2020b).
- The relative share of natural gas designated for electricity production out of the total natural gas was calculated.
- To determine the amount of methane emissions from gas fields due to natural gas used for electricity production, the relative share of natural gas designated for electricity production was multiplied by the total methane emissions.
- The quantity of electricity generated from natural gas was obtained from the CBS (2020a).
- The emissions were divided by the total electricity generated from natural gas to derive the EF per kWh.

Natural Gas Transportation:

- Methane emissions from natural gas transportation were calculated by multiplying the quantity of transported natural gas by an emission coefficient from the literature (Bradbury et al., 2015).
- The relative share of natural gas designated for electricity production out of the total natural gas was calculated.
- To determine the amount of methane emissions from the transportation system due to natural gas used for electricity production, the relative share of natural gas designated for electricity production was multiplied by the total methane emissions.
- The quantity of electricity generated from natural gas was obtained from the CBS (2020a).
- The emissions were divided by the total electricity generated from natural gas to derive the EF per kWh.

Ultimately, both factors were summed together, resulting in the upstream EF for natural gas.

National-grid and IPP-Specific Emission Factors

To obtain an average EF for the grid-electricity, a weighted average of both direct and upstream EFs was calculated, weighted by the generation output of each power station, including renewable sources. The resulting EFs of grid-electricity were 0.00052033 tCO_2eq/kWh (direct) and 0.00003878 tCO_2eq/kWh (upstream).

For Independent Power Producers (IPPs), each station's specific EFs (both direct and upstream EFs) were used without averaging, and these factors were applied to the relevant electricity consumption data to estimate GHG emissions. For reporting purposes, and due to confidentiality considerations, the average direct EF across IPPs was 0.00039051 (tCO_2eq/kWh), and the average upstream EF was 0.00005730 (tCO_2eq/kWh). In the analysis, however, the city-level calculations were based on the actual provider-specific EFs. Due to confidentiality constraints, these individual factors are not disclosed here but can be provided upon reasonable request to the corresponding author.

Supplementary Note 2. Extended details for transportation footprint calculations

The transportation footprint calculates the GHG emissions associated with city residents' travel. The emissions were calculated using EFs applied to vehicle mileage data. Direct fuel combustion emission factors were computed using the Israeli vehicle exhaust emissions database (MOEP, 2022a), which incorporates the country's vehicle inventory, traffic patterns, and climatic conditions. 'Well-to-tank' (WTT) emission factors were derived using Israeli fuel production data (MOEP, 2021) and vehicle fuel efficiency rates (VCA, 2023). All EFs were computed explicitly for this study as part of the framework. This footprint includes private transport emissions, adhering to the methodology outlined by Reznik et al. (2019), and public bus emissions estimated through a distinct method.

Private Transport

Private transport emissions were calculated following the methodology of Reznik et al. (2019). Vehicle-level data were obtained from the national vehicle registry maintained by the Ministry of Transportation (MOT, 2023a). The registry includes detailed attributes of all Israeli vehicles, such as vehicle type, engine size, ownership status, owner's city, and mileage records from mandatory roadworthiness check-ups..

Annual mileage estimation: The total distance traveled for each car in the analyzed cities was estimated based on the roadworthiness check-ups. Only vehicles that passed at least one inspection in 2019 were considered. For each car, the traveled distance (km) during 2019 was estimated under the assumption that the travel rate is uniform throughout the 365 days, and equal to the average rate based on test(s) conducted during 2019. Annual travel distances above the 75% quantile for taxis (86,088 km) were considered as data entry errors and excluded.

Emission factor matching: Each vehicle record was matched with the appropriate EF according to its characteristics (fuel type, engine size, emissions standard). Emissions were then estimated by applying 76 distinct direct fuel combustion factors and 11 upstream factors, all tailored to the vehicle's specifications. **Supplementary Table 4 includes all emission factors, categorized by fuel type, engine size, and emissions standards**.

Handling missing or inconsistent data: For privately owned cars with incomplete or inconsistent records (due to incorrectly typed or outlier test data, unidentified car model, etc), emissions were imputed using the median values of vehicles belonging to the same type ('motorcycle', 'car',

'other'), city, age category, and engine volume category. Only medians calculated from at least five vehicles were used for this purpose. Lastly, emissions totals were then aggregated across all privately owned cars for each of the 92 cities.

Leased vehicles: Leased vehicles required a separate procedure because they typically are not registered in the town of residence of the driver, but in that of the leasing company, making the association between leasing car test records and the city unknown. For these cars, emission estimates of all leasing vehicles nationwide were averaged to obtain country-level (of leasing vehicles only) averages of direct and indirect emissions. This averages were then scaled to the city level using the estimated share of leased vehicles in each city, derived from the annual household expenditure survey data conducted by the Central Bureau of Statistics (CBS, 2023).

Public transport

Public bus emissions were calculated using the Public Bus Lines dataset provided by the Ministry of Transportation (MOT, 2023b). This dataset contains detailed information on all bus lines, stations, route types, vehicle types, and trip frequencies nationwide. Annual mileage for each bus line was calculated by multiplying its weekly mileage by 52. Emissions were then estimated by applying emission factors (EFs) specific to vehicle type (bus or minibus), route type (inner-city, intercity, or regional), and traffic conditions (stop-and-go or heavy). The resulting emissions for each route and sub-route were attributed to the originating city, under the assumption that routes primarily serve the city's residents. **Supplementary Table 5 lists the full set of applied emission factors**.

Alternative method for cities not covered in the dataset: For three towns not included in the main dataset (Abu Ghosh, Kefar Tavor, and Pardesiyya), an alternative GIS-based method was used. This method relied on General Transit Feed Specification (GTFS) data from the Ministry of Transportation (MOT, 2022), which provides comprehensive licensing information on operators, bus lines, schedules, station sequences, and route paths. GTFS data were processed into a polyline GIS layer that included daily travel routes and weekly bus frequencies. Route lengths were calculated using ArcGIS Pro, and weekly mileage was estimated by multiplying route lengths by weekly frequencies. Since GTFS data reflect a single week, annual mileage was obtained by multiplying by 52, with minor adjustments for holidays and non-operational days.

Emissions were then calculated by multiplying the annual mileage by average weighted EFs for buses (Direct: 0.00094491 tCO_2eq/km; Well-to-tank: 0.00029424 tCO_2eq/km). Unlike the primary method (detailed in previous paragraph), specific route types were unavailable for these towns; therefore, average weighted EFs were applied to avoid systematic over- or underestimation. As in the main approach, resulting emissions for each route and sub-route were attributed to the city of origin, assuming the routes primarily serve local residents.

Emission factors calculation procedure

Direct emission factors

Direct emissions, resulting from vehicle operation, consist of hot and cold emissions:

1. Hot Emissions: Emissions released from the vehicle's exhaust once the engine reaches a stable operating temperature.
2. Cold Emissions: Excess emissions released until the engine reaches stable operating conditions. Note that cold emissions are added to hot emissions for the same travel distance, representing the surplus generated during cold starts.

Processing of hot and cold emissions data

Hot and cold emission factors were derived from the Israeli Vehicle Exhaust Emissions Database (MOEP, 2022a). These were calculated using the HBEFA 3.2 model, adapted to Israel's fleet composition, climate, traffic patterns, road types, inclines, and speed limits. Factors were initially reported in $kgCO_2$/km, $kgCH_4$/km, and kgN_2O/km, and were then converted to CO_2-equivalent units (tCO_2eq/km) using AR5 GWP values (CO_2 = 1, CH_4 = 28, N_2O = 265). Once both hot and cold EFs were processed, the following steps were taken to integrate them into a single weighted emission factor for each vehicle type:

Hot and cold emission factors were derived from the Israeli vehicle exhaust emissions database, developed by the Ministry of Environmental Protection (MOEP, 2022a). These factors were calculated using the HBEFA 3.2 model, which was adapted to Israel's specific conditions, including the vehicle fleet inventory, climate, traffic patterns, road types, inclines, and speed limits. The factors were originally provided in $kgCO_2$/km, $kgCH_4$/km, and kgN_2O/km, and were then converted to CO_2-equivalent units (tCO_2eq/km) using AR5 GWP values (CO_2 = 1, CH_4=28, N_2O=265). Once bith hot and cold EFs were processed, they were integrated into a single weighted EF for each vehicle type by applying the following steps:

1. The ratio of cold to hot emissions was first calculated for each vehicle type by dividing the cold EF by the corresponding hot EF.
2. For each vehicle type, the average annual mileage was obtained from the Central Bureau of Statistics (CBS, 2020c) for the year 2019. According to the Ministry of Environmental Protection (MOEP, 2022a), the average length of a cold-start trip in Israel is 2 km.

Also, assuming that a vehicle performs four cold starts per day (MOF, 2023), the total annual distance driven under cold-start conditions was calculated. This distance was then expressed as a percentage of the total annual mileage for the vehicle type, producing a factor that represents the share of kilometers driven under cold start conditions.

3. A Cold Emissions Additive Factor was then derived for each vehicle type by multiplying the cold-to-hot emissions ratio (from Step 1) by the cold-start mileage share (from Step 2).
4. Finally, the direct EF for each vehicle type was calculated by adding the product of the hot EF and the Cold Emissions Additive Factor to the hot EF itself:

$$EF_{direct} = EF_{hot} + \left(EF_{hot} \times \text{Cold emissions additive factor}\right)$$

Indirect emission factors

Indirect (well-to-tank, WTT) emission factors capture upstream GHG emissions from crude oil extraction, domestic refining, and imported fuel refining. The calculation procedure was:

1. **Emissions from crude oil extraction**: Emission factors (EFs) for GHG emissions from crude oil extraction were obtained from the Ministry of Environmental Protection's project on the external environmental costs of road transportation in Israel (MOEP, 2021). These factors, originally reported in $kgCO_2$/liter, $kgCH_4$/liter, and kgN_2O/liter, were converted to CO_2-equivalent units (tCO_2eq/liter) using the AR5 GWP values (CO_2 = 1, CH_4=28, N_2O=265). The resulting EF is identical for gasoline and diesel, as crude oil extraction occurs prior to refining, before the fuel types are and separated into by-products.
2. **Emissions from fuel refining**: Data on GHG emissions and fuel quantities for both domestically refined and imported fuels were obtained from MOEP (2021). For both diesel and gasoline, the data included (i) the amounts of domestically refined fuels (tonnes) and their associated GHG emissions (tCO_2eq), and (ii) the quantities of imported fuels and their corresponding GHG emissions. These values were used to calculate weighted emission factors for diesel and gasoline, with weights determined by fuel quantities. The resulting weighted EFs were initially expressed in tCO_2eq/tonne. Using conversion factors published by the Israeli Ministry of Energy (MOE, 2015), these were converted into tCO_2eq/liter, yielding final weighted emission factors for each fuel type.

3. **Calculation of total EFs per liter (tCO_2eq/liter)**: The total indirect emission factors for gasoline and diesel were calculated by summing each fuel's respective contributions from crude oil extraction (Step 1) and oil refining (Step 2), expressed in tCO_2eq/liter.

4. **Calculation of EFs per kilometer (tCO_2eq/km)**: Finally, the total EFs derived in Step 3 were combined with fuel efficiency data (liters/km), specific to vehicle type, fuel type, and engine size. This produced vehicle-specific EFs expressed in tCO_2eq/km. For electric vehicles, efficiency data (kWh/km) were multiplied by the previously calculated electricity EF (tCO_2eq/kWh), yielding the corresponding per-kilometer emission factor.

Fuel efficiency data was sourced from multiple sources:

- Passenger cars: Data was obtained from the Vehicle Certification Agency (VCA, 2023). Comprehensive datasets of passenger cars were downloaded, including details on fuel type, engine size, and other specifications, together with their corresponding fuel efficiency values, calculated according to WLTP combined range standard. Data from multiple years were included to ensure broad coverage, then processed and consolidated into a single large database. Duplicate entries and missing values (NA) were handled, after which the mean fuel efficiency (liters/km) was calculated for each fuel type and engine size category.
- Motorcycles, buses, and LPG-powered passenger vehicles: Data were taken from official published works of the MOEP (2015, 2021).
- Waste refuse and transfer trucks: Data were adapted from Franco González et al. (2021), a study focused on fuel-efficient driving in the context of urban waste collection.
- Electric vehicles: According to an official report jointly published by the Israeli Ministry of Finance and Ministry of Transportation (MOF, 2023), the electricity consumption of an EV in Israel is 0.17 ($\frac{kWh}{km}$).

Clarification on the calculation of emission factors for electric vehicles

Since electric vehicles (EVs) have no tailpipe emissions (EPA, 2023), the emissions considered in this study are those associated with the electricity required for their operation. Accordingly, two separate emission factors per vehicle-kilometre were calculated and subsequently applied in the analysis:

(1) Emissions associated with electricity generation

- **Electricity generation EF** – Calculated specifically for Israel using primary data from the 2019 national electricity grid mix (detailed in Supplementary Note 1). The resulting value was 0.000520327 ($\frac{tCO_2eq}{kWh}$).
- **Vehicle efficiency of a typical EV in Israel** – 0.17 ($\frac{kWh}{km}$), based on an official report published by the Israeli Ministry of Finance and Ministry of Transportation (MOF, 2023).

Multiplying these values gives:

$$0.000520327 \times 0.17 = 0.00008846 \left(\frac{tCO_2eq}{km}\right)$$

(2) Upstream emissions associated with the electricity supply-chain

- **Upstream EF for electricity generation**: Calculated using local primary data (detailed in Supplementary Note 1). The resulting value was 0.000038784 ($\frac{tCO_2eq}{kWh}$).
- **Typical electricity consumption of an EV in Israel** – 0.17 ($\frac{kWh}{km}$) (MOF, 2023).

Multiplying these values gives:

$$0.000038784 \times 0.17 = 0.00000659 \left(\frac{tCO_2eq}{km}\right)$$

Ultimately, the two factors were summed to produce a total electricity-related EF for EVs, which was then used in the framework's calculations of EV emissions.

Supplementary Note 3. Extended details for consumables footprint calculations

The consumables footprint accounts for the cradle-to-gate upstream emissions associated with selected short-lived materials consumed in urban areas, including paper, cardboard, plastic, metal, glass, and diapers. Given the lack of detailed consumption data, waste generation serves as a proxy for urban material consumption. This method aligns with the solid waste Life Cycle Assessment (LCA) approach, which presumes that most consumables, particularly short-term items, ultimately enter the waste stream (Kissinger et al., 2013). The following sections detail the procedures used to assess material quantities and calculate their associated emissions.

PART 1: Assessing consumed material quantities

Calculating landfilled consumable quantities

First, landfilled waste quantities of each city were obtained from the Local Authorities Waste and Recycling Survey (CBS, 2022). Waste composition data were taken from the National Waste Survey (MOEP, 2014), which provides detailed information on the composition of urban waste across cities. The amounts of specific consumables in each city were estimated by multiplying each city's total generated waste (landfilled and recycled) by the percentage share of each material type in the waste composition of each city.

Calculating recycled consumable quantities

Recycled quantities of consumables for each city were also obtained from CBS (2022). Exact data were available for 15 cities, representing 62% of the study population. For the remaining cities, where detailed data were not disclosed due to confidentiality, aggregate district-level data were used. aggregated data at the district level were used. District-level per capita recycling rates for each consumable were estimated and then multiplied by the population size of each city, resulting with the recycled quantities of each consumable. Note that in order to avoid overestimation and ensure accuracy, the 15 cities with detailed recycling data were excluded from the district-level averages estimations.

PART 2: Calculating GHG emissions

For each consumable type (i), the following steps were performed:

1. **GHG emissions from landfilled consumables:** Using the landfilled quantities from Part 1, emissions were calculated by multiplying each material's landfilled amount by its respective cradle-to-gate LCA emission factor (DEFRA, 2023; Cordella et al., 2015; Kissinger et al., 2013).

$$GHG_{\text{landfilled},i} = Q_{\text{landfilled},i} \times EF_{\text{LCA},i}$$

2. **Potential GHG emissions avoided by recycling**: Using the recycled quantities from Part 1, the avoided GHG emissions were estimated by multiplying the recycled amount of each consumable by its cradle-to-gate LCA factor. This calculation reflects the potential GHG emissions that would have occurred if the material had instead been produced from virgin sources.

$$GHG_{\text{potential},i} = Q_{\text{recycled},i} \times EF_{\text{LCA},i}$$

3. **Avoided emissions from recycling**:

   - **Calculating custom net emission Factors**: For each consumable, a custom net emission factor was calculated by taking the LCA factor for producing the material from virgin sources and subtracting it with the gross emission factor for recycling that material, taken from Turner et al. (2015). The result is the custom net factor, represents the emissions saving achieved trough recycling.

$$EF_{\text{net},i} = EF_{\text{LCA},i} - EF_{\text{gross},i}$$

   - This factor represents the difference in emissions between producing a material from virgin sources and recycling that material, capturing the emissions savings achieved through recycling. If the resulting value is negative, it indicates a complete offset of emissions, and it is treated as an absolute value in calculations.
   - **Calculating avoided emissions**: Once the custom net emission factor was obtained, it was multiplied by the recycled amount of each consumable, yielding the total avoided emissions (AE) due to recycling for that material.

$$AE_i = Q_{\text{recycled},i} \times EF_{\text{net},i}$$

4. **Net carbon footprint per consumable**: The avoided emissions are subtracted from the sum of GHG emissions from landfilled quantities and potential emissions from recycling, yielding the net carbon footprint for each consumable.

$$GHG_{\text{net},i} = (GHG_{\text{landfilled},i} + GHG_{\text{potential},i}) - AE_i$$

5. **Total net carbon footprint:** Finally, the net carbon footprints of all consumables were summed to obtain the city-level total net carbon footprint.

$$GHG_{\text{total,net}} = \sum GHG_{\text{net},i}$$

The calculated emission factors are presented in Supplementary Table 6.

Supplementary Note 4. Extended details for waste footprint calculations

The waste footprint covers the direct and downstream emissions from waste management and comprises three components: collection, transfer, and landfilling. The collection and transfer emissions pertain to the trucks involved in the initial waste collection and their following transferring to transit and landfill sites. The landfilling component accounts for the GHGs generated from the disposal process.

Collection

Emissions of the collection phase were estimated by initially calculating the annual mileage of refuse trucks through a GIS-based analysis of urban road networks. The analysis focused on identifying the streets that pass through residential areas and public-commercial institutions, while excluding industrial and agricultural zones. The exclusion was necessary as the study concentrated on Municipal Solid Waste (MSW), which includes commercial and household waste but not industrial waste.

To calculate the total distance traveled by refuse trucks within each city, GIS tools were used to intersect layers of roads and land use. The length of relevant roads in each city was identified, and its distance was multiplied by 3, reflecting a typical weekly collection frequency in Israeli municipalities (State Comptroller of Israel, 2022). The resulting weekly mileage was then multiplied by 52 to estimate the total annual mileage.

Next, the total annual mileage was multiplied by fuel combustion and WTT emission factors (tCO_2eq/km). The factors were calculated using the same methodology detailed in the transportation section. The EFs were formulated explicitly for a typical Israeli refuse truck, a 26-ton compactor truck commonly used by Israeli municipalities for waste collection, and were designed to reflect the "stop-and-go" driving pattern of refuse trucks (Daskal, 2018). The direct and WTT EFs are 0.00160061 (tCO_2eq/km) and 0.00020628 (tCO_2eq/km), respectively.

Transfer

Emissions from the waste-transfer stage were estimated by calculating the annual mileage of trucks transporting waste to transit stations and landfill sites. First, each city's relevant destination points—whether transit stations or landfill sites—were identified through direct communication with municipal waste administrators, and the sites were geographically mapped. Next, trip distances between cities and stations, as well as between stations and landfills, were measured using Google Maps. Given that transfer routes differ among cities—some transport waste directly to landfills, while others use transit stations before reaching the landfill—three travel types were defined, each characterized by distinct trip segments:

- **Type 1**: The city transports its waste using a refuse truck carrying 10 tons from the city to a transit station outside the city's boundaries. This segment includes round-trip trips since refuse trucks return to the city. The waste is then transported from the transit station to the landfill site using full-trailer trucks carrying 35 tons of waste. This segment includes only one-way trips since the full-trailer trucks do not return to the city.
- **Type 2**: In this case, the transit station is located within the city's boundaries. Therefore, only the segment from the city's transit station to the landfill site is calculated using full-trailer trucks carrying 35 tons of waste (one-way only).
- **Type 3**: In this case, the waste does not pass through a transit station and is transported directly from the city to the landfill site using refuse trucks carrying 10 tons of waste. This segment includes round-trip trips as the refuse trucks return to the city.

Each city was assigned its appropriate travel type and relevant trip segments. Trip frequencies in each segment were determined by dividing the collected waste quantities (CBS, 2022) by truck capacities (Daskal, 2018), assuming trucks are fully loaded (due to the lack of specific data on actual load per trip). These frequencies, combined with the initially calculated trip distances, were used to calculate the total mileage for each segment.

The resulting mileage was converted into GHG emissions by applying direct and WTT emission factors, each specifically tailored to the truck types used in each segment. For 26-ton refuse trucks, the direct and indirect factors were 0.00100711 and 0.00017294 tCO_2eq/km, respectively. For 35-ton full-trailer lorries, the corresponding values were 0.00126392 and 0.00022495 tCO_2eq/km; Ultimately, the total transfer emissions for each city were determined

by summing the emissions across all relevant trip segments.

Landfilling

Landfilling emissions were estimated using the GPC's Methane Commitment method (WRI et al., 2021), which employs a lifecycle and mass-balance approach. This method primarily relies on the annual mass of municipal solid waste (MSW) sent to landfills and its composition. Emissions calculations were performed using the IPCC Methane Commitment formula (Eq. 1), based on MSW mass data from the local authorities' waste survey (CBS, 2022). The estimation included various types of MSW collected by municipalities to assess the degradable organic carbon (DOC) content (Eq. 2).

Eq 1:

$$\text{CH}_4 \text{ emissions} = MSW_T \times MCF \times DOC \times DOC_F \times F \times \frac{16}{12} \times (1 - f_{\text{rec}}) \times (1 - OX)$$

Where:

- MSW_T is mass of solid waste sent to landfill in inventory year
- MCF is methane correction factor (1.0, according to MOEP, 2015)
- DOC is degradable organic carbon in year of deposition
- DOC_F is fraction of DOC that is ultimately degraded (0.6, the IPCC default)
- F is fraction of methane on landfill gas (0.5, the IPCC default)
- 16/12 is Stoichiometric ratio between methane and carbon
- f_{rec} is fraction of methane recovered (0.33, according to MOEP, 2018)
- OX is oxidation factor (0.1 for well-managed landfills)

Eq 2:

$$DOC = (0.15 \times A) + (0.2 \times B) + (0.4 \times C) + (0.43 \times D) + (0.24 \times E) + (0.15 \times F)$$

Where: A to F represent the fractions of various waste components (food, garden waste, paper, wood, textiles, and industrial waste, respectively). Each city's waste composition was sourced from the national waste survey (MOEP, 2014). For a city not featured in the survey, the composition of a city with similar socio-economic and demographic characteristics was applied.

Supplementary Note 5. Extended details for water footprint calculations

The water footprint measures the upstream emissions associated with urban water consumption. Emissions were calculated by applying an LCA-based EF to each city's water consumption quantities (m^3) reported by the Israeli Water Authority (NWA, 2020a). This report details water consumption values of each city and town in the country, covering residential, public services, commercial, and healthcare services, and infrastructure water loss.

The EF is adapted from the work of Meron et al. (2020), a life cycle assessment (LCA) study that assesses the environmental impacts associated with the provision of tap water by the Israeli supply system. The EF accounts for the GHGs associated with abstraction, treatment, distribution, and leakage throughout the water supply chain. Originally set for 2015, the EF was adapted to reflect the 2019 electricity fuel mix, resulting in 0.0016867 (tCO_2eq/m^3). The following steps outline the calculation steps carried out to derive this value:

1. **Calculate the Base Electricity Emission Factor for 2015:** According to Meron et al. (2020): The total EF for water supply in 2015 is 2.27 ($kgCO_2eq/m^3$), with electricity contributing 89% of this total. The impact score of electricity in 2015 is 7.86E-01 ($kgCO_2eq/kWh$). Thus, the EF due to electricity is calculated as:

$$\mathrm{EF}_{\mathrm{electricity},2015} = 2.27 \left(\frac{\mathrm{kgCO_2eq}}{\mathrm{m}^3}\right) \times 0.89 = 2.0203 \left(\frac{\mathrm{kgCO_2eq}}{\mathrm{m}^3}\right)$$

2. **Determine Electricity Consumption per Cubic Meter for 2015:** The electricity consumption per cubic meter is determined by dividing the EF from electricity (step 1) by the impact score of electricity in 2015 (Meron et al., 2020):

$$\begin{aligned}\text{Electricity per m}^3 &= \frac{\mathrm{EF}_{\mathrm{electricity},2015}}{\text{Impact score of electricity (2015)}} = \frac{2.0203 \left(\frac{\mathrm{kgCO_2eq}}{\mathrm{m}^3}\right)}{7.86 \times 10^{-1} \left(\frac{\mathrm{kgCO_2eq}}{\mathrm{kWh}}\right)} \\ &\approx 2.57 \left(\frac{\mathrm{kWh}}{\mathrm{m}^3}\right)\end{aligned}$$

3. **Adjust The Electricity EF for 2019:** The national EF of electricity consumption in 2019 (0.55911 $kgCO_2eq/kWh$)) is applied to the electricity consumption per cubic meter (step 2):

$$EF_{electricity,2019} = 2.57 \left(\frac{kWh}{m^3}\right) \times 0.55911 \left(\frac{kgCO_2eq}{kWh}\right) \approx 1.437 \left(\frac{kgCO_2eq}{m^3}\right)$$

4. **Account for Non-Electricity Emissions:** The EF of non-electricity sources was assumed to remain the same between 2015 and 2019 due to the lack of available information suggesting otherwise:

$$EF_{other} = 2.27 \left(\frac{kgCO_2eq}{m^3}\right) \times (1 - 0.89) \approx 0.2497 \left(\frac{kgCO_2eq}{kWh}\right)$$

5. **Calculate the Total EF for Water Supply in 2019:** The total EF is obtained by summing the adjusted electricity EF (step 3) with the non-electricity EF (step 4):

$$EF_{total,2019} = 1.437 \left(\frac{kgCO_2eq}{m^3}\right) + 0.2497 \left(\frac{kgCO_2eq}{kWh}\right) \approx 1.6867 \left(\frac{kgCO_2eq}{m^3}\right)$$

Supplementary Note 6. Extended details for wastewater footprint calculations

The wastewater footprint quantifies the GHG emissions from each city's wastewater treatment. The primary data source used was the Israeli Wastewater Management Survey (NWA, 2020b), which details each city's wastewater quantities, treatment locations, treatment methods, and effluent quality.The analysis applied 68 facility-specific emission factors (EFs), each tailored to the wastewater treatment plants (WWTPs) serving the analyzed cities. These factors, developed as part of the presented framework, account for both direct on-site treatment-related emissions and indirect energy-related emissions. They were derived using the survey data (NWA, 2020b), relevant local literature (Fine & Hadas, 2012), and the Ministry of Environmental Protection's wastewater GHG calculators (MOEP, 2022b). The following paragraphs describe the procedure for calculating these factors. **The EFs for each WWTP are provided in Supplementary Table 7.**

Emission factors calculation procedure

In total, EFs were calculated for 68 different WWTPs serving the analyzed cities. The process began with data collection for each facility, sourced from the Israeli Wastewater Survey (NWA, 2020b). The data included WWTP name, total treated wastewater volumes (including wastewater from cities not analyzed in this study), treatment type, population size served by the WWTP, and sample data (average COD concentration in mg/liter).

Direct emission factors

- For each WWTP, total GHG emissions from the treatment process were calculated using the official wastewater emission calculator of the Israeli Ministry of Environmental Protection (MOEP, 2022b).
- The direct EF was obtained by dividing total GHG emissions by the total wastewater volume treated.

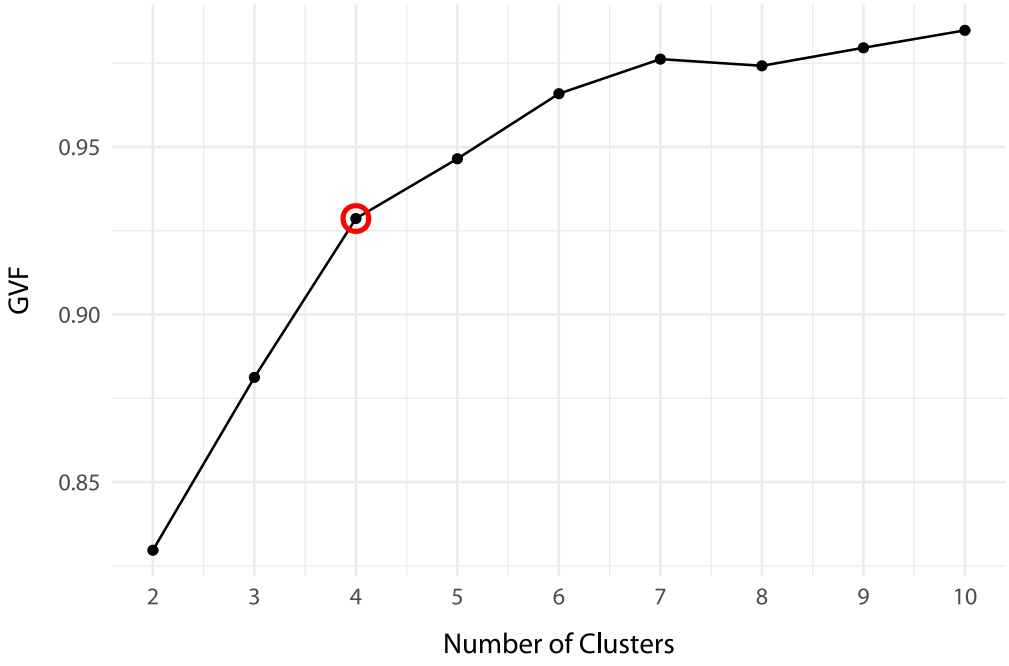
Energy-related emission factors

- As facility-specific electricity consumption data were unavailable, electricity consumption factors from the literature were used (Fine & Hadas, 2012). These factors are based on the volume of treated wastewater (kWh/m^3).
- For each WWTP, the treated volume was multiplied by the corresponding electricity consumption factor to estimate total electricity consumption (kWh).
- Total electricity consumption was multiplied by Israel-specific emission factors for electricity generation, covering both power generation and upstream processes (Same as the ones provided in Supplementary Note 1), to derive total electricity-related emissions of each WWTP.
- Lastly, the indirect EF for each WWTP was calculated by dividing these emissions by the total volume of wastewater treated.

Supplementary Note 7. Clustering and relative deviation calculation

Clustering

Following the emissions assessments, the cities were segmented into clusters based on their total per-capita CFs to identify patterns and groupings in urban emissions. The Jenks natural breaks optimization method was employed to determine optimal breakpoints in the data, maximizing homogeneity within clusters while ensuring heterogeneity between them (Jenks, 1967); To determine the optimal number of clusters, the Goodness of Variance Fit (GVF) was calculated for different cluster counts. This measure helps identify how well the data is partitioned, with higher values indicating better clustering. The GVF peaked at 0.93 with four clusters, indicating the optimal number of clusters. Although higher cluster counts could slightly increase GVF, they would result in overly fragmented data, reducing the practical utility of the clusters. Therefore, four clusters were selected to balance detail and usability. The clusters were defined by the following breakpoints: 1.76, 3.41, 4.75, 5.90, and 7.28 tCO_2eq/cap. These breakpoints reflect ascending per-capita emissions in higher clusters, indicating escalating mitigation urgency.



Goodness of Variance Fit (GVF) by number of clusters

Relative Deviation

This analysis calculates the relative deviation of each city's per-capita emissions from the national mean for each footprint category/subcategory, providing a benchmarked perspective on urban emissions. The approach identifies significant deviations, highlighting how each city's emissions compare to the national average and facilitating effective benchmarking among cities. The relative deviation is expressed as a percentage, calculated using the formula:

$$D_j^k = \frac{F_j^k / P_j}{F^k / P} - 1$$

Where:

- D_j^k represents the relative deviation for city j in footprint category k
- F_j^k is the total emissions for city j in footprint category k
- P_j is the population of city j
- F^k is the national total emissions in footprint category k
- P is the national population

Supplementary Tables

Supplementary Table 1. List of 92 analyzed cities with population, area, district, municipal type, and socio-economic cluster (2019).

City	Population	Area	District	Type	SE
Abu Ghosh	7,698	1.89	Jerusalem	Local Council	3
Afula	54,250	29.17	North	Municipality	5
Ar'ara	25,245	9.18	Haifa	Local Council	3
Ar'ara-BaNegev	18,332	14.03	South	Local Council	1
Arad	26,756	126.13	South	Municipality	3
Ashdod	225,939	63.90	South	Municipality	5
Ashqelon	144,073	52.30	South	Municipality	5
Azor	12,957	2.22	Tel Aviv	Local Council	7
Bat Yam	129,013	9.40	Tel Aviv	Municipality	5
Be'er Sheva	209,687	117.40	South	Municipality	5
Be'er Ya'aqov	27,768	9.46	Central	Local Council	7
Beit Jann	12,001	7.20	North	Local Council	4
Bene Beraq	204,639	7.30	Tel Aviv	Municipality	2
Bet She'an	18,464	7.39	North	Municipality	5
Bet Shemesh	124,957	38.30	Jerusalem	Municipality	2
Bir El-Maksur	9,522	5.83	North	Local Council	2
Daliyat Al-Karmel	17,653	9.44	Haifa	Local Council	5
Dimona	34,500	220.46	South	Municipality	5
El'ad	48,763	3.49	Central	Municipality	2
Even Yehuda	13,826	8.27	Central	Local Council	9
Giv'atayim	60,644	3.24	Tel Aviv	Municipality	9
Hadera	97,335	56.28	Haifa	Municipality	6
Haifa	285,316	72.90	Haifa	Municipality	7
Hazor HaGelilit	9,272	5.46	North	Local Council	4
Herzliyya	97,470	24.07	Tel Aviv	Municipality	8
Hod HaSharon	63,175	19.27	Central	Municipality	9
Holon	196,282	19.00	Tel Aviv	Municipality	7
Hura	22,337	8.74	South	Local Council	1
I'billin	13,305	12.08	North	Local Council	3
Jaljulye	10,148	1.89	Central	Local Council	3
Jerusalem	936,425	125.50	Jerusalem	Municipality	2
Jisr Az-Zarqa	14,930	2.02	Haifa	Local Council	2
Kafar Bara	3,678	2.28	Central	Local Council	4

Table 1 continued from previous page

City	Population	Area	District	Type	SE
Kafar Kama	3,373	8.83	North	Local Council	6
Kafar Kanna	22,751	10.06	North	Local Council	2
Kefar Sava	101,432	14.50	Central	Municipality	8
Kefar Tavor	4,311	12.39	North	Local Council	8
Kokhav Ya'ir	8,702	3.05	Central	Local Council	10
Laqye	14,075	6.86	South	Local Council	1
Lehavim	6,415	11.15	South	Local Council	10
Lod	77,223	12.09	Central	Municipality	4
Majd al-Kurum	15,455	9.25	North	Local Council	3
Mazkeret Batya	14,837	7.42	Central	Local Council	8
Metar	9,418	38.82	South	Local Council	9
Mevasseret Ziyyon	24,052	6.28	Jerusalem	Local Council	8
Mizpe Ramon	5,211	75.84	South	Local Council	3
Modi'in-Makkabbim-Re'ut	93,277	48.33	Central	Municipality	9
Nahariyya	58,096	13.78	North	Municipality	6
Nazareth	77,445	14.17	North	Municipality	3
Nes Ziyyona	50,351	15.68	Central	Municipality	8
Nesher	24,148	12.94	Haifa	Municipality	7
Netanya	221,353	34.70	Central	Municipality	6
Nof HaGalil	41,734	32.87	North	Municipality	5
Omer	7,530	18.22	South	Local Council	10
Or Aqiva	18,972	5.55	Haifa	Municipality	5
Or Yehuda	36,813	6.66	Tel Aviv	Municipality	5
Pardes Hanna-Karkur	43,011	22.84	Haifa	Local Council	7
Pardesiyya	5,930	1.47	Central	Local Council	9
Petah Tiqwa	247,956	35.60	Central	Municipality	7
Qalansawe	23,143	8.34	Central	Municipality	3
Qazrin	7,297	10.89	North	Local Council	5
Qiryat Atta	59,030	17.34	Haifa	Municipality	6
Qiryat Gat	57,105	15.84	South	Municipality	4
Qiryat Motzkin	43,303	3.84	Haifa	Municipality	7
Qiryat Ono	40,409	4.52	Tel Aviv	Municipality	9
Qiryat Shemona	22,512	14.38	North	Municipality	5
Qiryat Tiv'on	18,130	8.25	Haifa	Local Council	8
Qiryat Ye'arim	6,128	0.50	Jerusalem	Local Council	2
Ra'annana	75,421	14.86	Central	Municipality	8

Table 1 continued from previous page

City	Population	Area	District	Type	SE
Rahat	71,437	33.49	South	Municipality	1
Ramat Gan	163,480	16.40	Tel Aviv	Municipality	8
Ramat HaSharon	47,245	16.64	Tel Aviv	Municipality	9
Ramat Yishay	7,785	2.33	North	Local Council	8
Ramla	76,246	11.99	Central	Municipality	5
Rehovot	143,904	23.80	Central	Municipality	7
Rekhasim	12,668	3.31	Haifa	Local Council	2
Rishon LeZiyyon	254,384	61.90	Central	Municipality	7
Rosh HaAyin	61,801	15.87	Central	Municipality	7
Sakhnin	31,702	11.53	North	Municipality	4
Sederot	27,635	6.31	South	Municipality	5
Segev-Shalom	10,936	13.52	South	Local Council	1
Shefar'am	42,137	19.58	North	Municipality	3
Shelomi	6,511	5.85	North	Local Council	6
Shoham	21,045	7.16	Central	Local Council	9
Tel Aviv - Yafo	460,613	57.20	Tel Aviv	Municipality	8
Tel Mond	13,144	7.13	Central	Local Council	9
Tur'an	14,240	11.91	North	Local Council	3
Umm al-Fahm	56,109	26.03	Haifa	Municipality	2
Yanuh-Jat	6,679	13.55	North	Local Council	5
Yeroham	10,294	58.60	South	Local Council	4
Yirka	17,171	15.63	North	Local Council	3
Yoqne'am Illit	23,796	8.31	North	Municipality	7

Supplementary Table 2. Per-capita carbon footprints of analyzed cities (tCO_2eq/cap)

ID	City name	Electricity	Transportation	Waste	Consumables	Water	Wastewater	**Total**
89	Abu Ghosh	1.25	1.03	0.49	0.40	0.10	0.05	3.32
30	Afula	2.58	1.25	0.54	0.52	0.13	0.06	5.07
60	Ar'ara	0.90	1.05	0.54	0.55	0.04	0.03	3.11
84	Ar'ara-BaNegev	0.33	0.70	0.28	0.27	0.15	0.03	1.76
49	Arad	1.55	1.70	0.56	0.53	0.12	0.12	4.58
6	Ashdod	2.50	0.97	0.61	0.66	0.12	0.13	5.00
12	Ashqelon	1.96	1.15	0.78	0.83	0.15	0.09	4.97
61	Azor	2.87	1.53	0.72	0.63	0.15	0.12	6.02
19	Bat Yam	1.86	0.88	0.45	0.44	0.11	0.09	3.83
7	Be'er Sheva	2.37	1.36	0.65	0.70	0.15	0.09	5.34
45	Be'er Ya'aqov	2.64	1.34	0.64	0.56	0.13	0.07	5.37
70	Beit Jann	1.45	1.10	0.72	0.72	0.11	0.06	4.18
14	Bene Beraq	1.77	0.46	0.36	0.46	0.09	0.08	3.20
48	Bet She'an	2.94	1.21	1.31	1.25	0.16	0.09	6.95
18	Bet Shemesh	1.67	0.75	0.63	0.80	0.09	0.05	3.99
82	Bir El-Maksur	1.51	1.18	0.35	0.36	0.14	0.07	3.61
57	Daliyat Al-Karmel	1.90	1.46	0.80	0.77	0.19	0.07	5.20
40	Dimona	2.12	1.48	0.77	0.75	0.13	0.15	5.40
47	El'ad	1.19	0.63	0.40	0.52	0.09	0.06	2.88
56	Even Yehuda	3.17	1.69	0.80	0.72	0.18	0.17	6.73
32	Giv'atayim	2.19	1.01	0.34	0.32	0.12	0.09	4.07
15	Hadera	2.68	1.28	0.75	0.81	0.14	0.07	5.72
3	Haifa	2.61	1.22	0.58	0.47	0.12	0.11	5.10
64	Hazor HaGelilit	3.47	1.31	1.09	0.93	0.15	0.09	7.05
13	Herzliyya	3.69	1.30	0.94	0.86	0.19	0.10	7.07
25	Hod HaSharon	2.67	1.47	0.60	0.54	0.14	0.12	5.53
9	Holon	2.22	1.13	0.48	0.41	0.12	0.09	4.46
79	Hura	0.40	0.68	0.29	0.29	0.12	0.05	1.82
68	I'billin	1.71	1.30	0.60	0.49	0.13	0.10	4.32
78	Jaljulye	1.41	1.42	0.54	0.45	0.16	0.06	4.04
1	Jerusalem	2.35	0.92	0.40	0.53	0.09	0.11	4.40
77	Jisr Az-Zarqa	1.00	0.84	0.42	0.42	0.13	0.05	2.87

Table 2 continued from previous page

ID	City name	Electricity	Transportation	Waste	Consumables	Water	Wastewater	**Total**
92	Kafar Bara	1.40	1.64	0.51	0.43	0.11	0.03	4.12
91	Kafar Kama	2.08	1.37	0.65	0.70	0.12	0.14	5.06
59	Kafar Kanna	1.51	1.19	0.34	0.34	0.12	0.17	3.66
16	Kefar Sava	2.53	1.32	0.48	0.44	0.13	0.10	5.00
85	Kefar Tavor	3.28	2.44	0.61	0.55	0.17	0.23	7.28
67	Kokhav Ya'ir	2.63	2.40	0.75	0.77	0.20	0.15	6.91
86	Laqye	0.50	1.03	0.21	0.21	0.16	0.08	2.18
83	Lehavim	2.09	1.81	0.49	0.51	0.19	0.10	5.19
17	Lod	3.77	1.07	0.79	0.75	0.11	0.06	6.55
65	Majd al-Kurum	1.48	1.16	0.64	0.64	0.11	0.09	4.11
62	Mazkeret Batya	1.81	1.62	0.55	0.51	0.12	0.08	4.69
73	Metar	1.66	1.74	0.63	0.56	0.20	0.11	4.89
51	Mevasseret Ziyyon	1.84	1.70	0.54	0.49	0.13	0.12	4.81
87	Mizpe Ramon	2.04	1.57	1.02	0.86	0.15	0.11	5.76
21	Modi'in-Makkabbim-Re'ut	2.12	1.59	0.38	0.35	0.13	0.08	4.65
24	Nahariyya	2.96	1.19	0.79	0.86	0.14	0.09	6.01
23	Nazareth	1.85	1.24	0.67	0.67	0.12	0.14	4.69
31	Nes Ziyyona	2.35	1.31	0.56	0.51	0.13	0.10	4.96
50	Nesher	2.51	1.20	0.57	0.49	0.12	0.10	4.99
8	Netanya	2.35	1.09	0.64	0.71	0.12	0.10	5.02
39	Nof HaGalil	1.99	1.37	0.46	0.45	0.11	0.17	4.55
69	Omer	3.23	2.13	0.61	0.63	0.22	0.09	6.91
52	Or Aqiva	2.89	1.39	0.69	0.66	0.17	0.08	5.89
38	Or Yehuda	2.49	1.21	0.64	0.63	0.12	0.10	5.20
34	Pardes Hanna-Karkur	2.37	1.59	0.45	0.38	0.15	0.07	5.02
81	Pardesiyya	2.41	2.02	0.58	0.52	0.18	0.12	5.83
4	Petah Tiqwa	2.71	1.11	0.68	0.57	0.13	0.10	5.31
55	Qalansawe	1.59	1.28	0.55	0.56	0.11	0.05	4.13
75	Qazrin	2.95	1.69	0.60	0.58	0.18	0.07	6.08
29	Qiryat Atta	2.28	1.26	0.51	0.56	0.12	0.10	4.82

Table 2 continued from previous page

ID	City name	Electricity	Transportation	Waste	Consumables	Water	Wastewater	**Total**
28	Qiryat Gat	2.55	1.05	0.62	0.58	0.12	0.10	5.01
36	Qiryat Motzkin	1.96	1.24	0.68	0.58	0.12	0.10	4.68
41	Qiryat Ono	2.07	1.21	0.38	0.35	0.12	0.10	4.24
43	Qiryat Shemona	3.85	1.48	0.75	0.72	0.14	0.23	7.18
58	Qiryat Tiv'on	2.25	1.63	0.45	0.41	0.14	0.17	5.05
90	Qiryat Ye'arim	1.21	0.77	0.48	0.62	0.10	0.04	3.22
20	Ra'annana	3.29	1.41	0.76	0.69	0.16	0.09	6.39
35	Rahat	0.70	0.97	0.57	0.44	0.10	0.07	2.85
10	Ramat Gan	3.21	1.05	0.44	0.41	0.13	0.10	5.35
27	Ramat HaSharon	3.75	1.31	0.59	0.54	0.17	0.12	6.47
71	Ramat Yishay	2.58	2.14	0.73	0.65	0.12	0.20	6.42
22	Ramla	2.55	1.16	0.70	0.69	0.11	0.07	5.28
11	Rehovot	3.03	1.20	0.73	0.61	0.12	0.09	5.78
76	Rekhasim	1.30	0.69	0.55	0.71	0.09	0.07	3.41
5	Rishon LeZiyyon	2.51	1.37	0.56	0.50	0.13	0.10	5.17
26	Rosh HaAyin	2.46	1.51	0.61	0.52	0.14	0.09	5.33
46	Sakhnin	1.64	1.35	0.67	0.67	0.11	0.09	4.52
42	Sederot	2.78	1.28	0.83	0.80	0.14	0.07	5.90
74	Segev-Shalom	2.16	0.67	0.51	0.49	0.17	0.16	4.16
37	Shefar'am	1.73	1.20	0.79	0.79	0.14	0.10	4.75
80	Shelomi	3.07	1.38	0.29	0.31	0.11	0.27	5.42
54	Shoham	2.51	1.64	0.35	0.32	0.16	0.09	5.08
2	Tel Aviv - Yafo	3.89	1.25	0.68	0.58	0.17	0.13	6.72
63	Tel Mond	2.29	1.63	0.50	0.45	0.16	0.16	5.20
66	Tur'an	1.81	1.53	0.37	0.31	0.11	0.18	4.31
33	Umm al-Fahm	1.32	1.24	0.65	0.55	0.13	0.10	3.99
88	Yanuh-Jat	1.43	1.18	0.79	0.76	0.12	0.14	4.41
72	Yeroham	1.85	1.40	0.67	0.57	0.15	0.14	4.78
53	Yirka	2.64	1.10	1.21	1.21	0.13	0.14	6.43
44	Yoqne'am Illit	3.77	1.48	0.60	0.51	0.13	0.16	6.64

Supplementary Table 3. Absolute carbon footprints of analyzed cities (tCO_2eq)

City	Electricity	Transportation	Waste	Consumables	Water	Wastewater	**Total**	#
Abu Ghosh	9,631	7,931	3,750	3,111	755	400	25,577	89
Afula	139,995	67,738	29,149	28,292	7,068	3,018	275,260	30
Ar'ara	22,604	26,508	13,729	13,865	1,128	768	78,602	60
Ar'ara-BaNegev	6,135	12,794	5,099	5,027	2,719	581	32,354	84
Arad	41,527	45,377	15,046	14,135	3,175	3,248	122,508	49
Ashdod	565,434	219,780	138,527	149,946	26,139	30,412	1,130,237	6
Ashqelon	282,050	165,957	113,024	119,845	21,550	13,411	715,837	12
Azor	37,128	19,792	9,389	8,103	1,980	1,597	77,990	61
Bat Yam	239,338	113,745	57,856	57,385	14,573	11,590	494,487	19
Be'er Sheva	496,927	286,137	136,504	147,400	32,452	19,760	1,119,180	7
Be'er Ya'aqov	73,212	37,313	17,804	15,465	3,498	1,875	149,167	45
Beit Jann	17,361	13,255	8,683	8,687	1,368	778	50,132	70
Bene Beraq	361,617	93,207	73,507	93,394	17,853	15,744	655,320	14
Bet She'an	54,284	22,289	24,115	23,136	2,955	1,571	128,350	48
Bet Shemesh	208,974	93,284	78,906	100,143	11,390	6,053	498,749	18
Bir El-Maksur	14,357	11,274	3,352	3,383	1,314	695	34,375	82
Daliyat Al-Karmel	33,566	25,844	14,044	13,627	3,399	1,281	91,762	57
Dimona	73,100	50,902	26,655	25,786	4,533	5,343	186,321	40
El'ad	57,907	30,710	19,527	25,130	4,155	3,062	140,489	47
Even Yehuda	43,894	23,334	11,089	9,929	2,507	2,328	93,081	56
Giv'atayim	133,074	61,283	20,787	19,442	7,005	5,253	246,844	32
Hadera	260,435	124,171	73,474	78,829	13,663	6,562	557,135	15
Haifa	744,045	348,477	164,179	133,069	33,432	32,088	1,455,291	3
Hazor HaGelilit	32,140	12,163	10,144	8,614	1,400	862	65,322	64
Herzliyya	359,185	126,693	92,069	83,484	18,282	9,832	689,546	13
Hod HaSharon	168,362	93,147	37,607	33,938	8,686	7,425	349,166	25
Holon	436,159	221,865	94,101	80,457	24,367	18,523	875,473	9
Hura	8,944	15,127	6,504	6,371	2,605	1,207	40,757	79
I'billin	22,702	17,271	7,932	6,479	1,699	1,373	57,457	68
Jaljulye	14,350	14,442	5,429	4,549	1,601	625	40,996	78
Jerusalem	2,202,874	862,206	374,559	494,320	83,717	98,727	4,116,402	1
Jisr Az-Zarqa	14,868	12,611	6,274	6,326	2,005	758	42,841	77

Table 3 continued from previous page

City	Electricity	Transportation	Waste	Consumables	Water	Wastewater	**Total**	#
Kafar Bara	5,159	6,037	1,870	1,568	389	113	15,137	92
Kafar Kama	7,025	4,612	2,184	2,364	416	468	17,070	91
Kafar Kanna	34,348	26,973	7,660	7,731	2,647	3,864	83,223	59
Kefar Sava	256,495	133,500	48,636	44,892	13,249	10,296	507,069	16
Kefar Tavor	14,144	10,538	2,615	2,368	719	1,002	31,386	85
Kokhav Ya'ir	22,905	20,922	6,505	6,723	1,757	1,339	60,151	67
Laqye	7,034	14,433	2,925	2,923	2,307	1,097	30,720	86
Lehavim	13,400	11,592	3,164	3,289	1,220	649	33,315	83
Lod	291,300	82,344	60,980	57,600	8,696	4,630	505,549	17
Majd al-Kurum	22,815	17,902	9,823	9,872	1,679	1,378	63,469	65
Mazkeret Batya	26,871	24,004	8,169	7,517	1,803	1,222	69,587	62
Metar	15,658	16,358	5,906	5,269	1,901	1,008	46,100	73
Mevasseret Ziyyon	44,235	40,835	12,947	11,872	3,081	2,817	115,786	51
Mizpe Ramon	10,645	8,173	5,331	4,494	758	594	29,996	87
Modi'in-Makkabbim-Re'ut	197,963	148,121	35,297	32,861	12,425	7,042	433,709	21
Nahariyya	171,740	68,932	45,985	49,703	7,931	5,090	349,381	24
Nazareth	143,312	95,911	51,929	52,077	9,539	10,605	363,372	23
Nes Ziyyona	118,516	65,709	28,243	25,926	6,324	5,010	249,728	31
Nesher	60,496	29,070	13,735	11,836	3,015	2,446	120,597	50
Netanya	519,613	241,891	142,281	156,473	27,637	22,279	1,110,173	8
Nof HaGalil	83,110	57,190	19,239	18,623	4,633	7,039	189,834	39
Omer	24,318	16,020	4,605	4,767	1,659	673	52,041	69
Or Aqiva	54,850	26,462	13,119	12,579	3,131	1,591	111,731	52
Or Yehuda	91,836	44,664	23,449	23,093	4,590	3,661	191,292	38
Pardes Hanna-Karkur	102,045	68,376	19,375	16,524	6,611	2,865	215,796	34
Pardesiyya	14,316	11,975	3,411	3,080	1,080	703	34,565	81
Petah Tiqwa	671,487	276,154	168,596	142,011	33,268	25,560	1,317,075	4
Qalansawe	36,806	29,604	12,688	12,915	2,520	1,073	95,606	55
Qazrin	21,548	12,315	4,399	4,246	1,298	534	44,340	75
Qiryat Atta	134,738	74,202	30,094	32,779	7,158	5,687	284,659	29

Table 3 continued from previous page

City	Electricity	Transportation	Waste	Consumables	Water	Wastewater	**Total**	#
Qiryat Gat	145,538	59,875	35,553	32,842	6,660	5,757	286,224	28
Qiryat Motzkin	85,089	53,637	29,329	25,115	5,037	4,425	202,632	36
Qiryat Ono	83,701	49,029	15,267	14,188	5,007	4,141	171,332	41
Qiryat Shemona	86,693	33,206	16,951	16,236	3,198	5,290	161,575	43
Qiryat Tiv'on	40,780	29,526	8,140	7,504	2,514	3,017	91,482	58
Qiryat Ye'arim	7,445	4,700	2,961	3,770	604	262	19,743	90
Ra'annana	247,777	106,347	57,247	52,094	12,280	6,427	482,171	20
Rahat	49,665	69,541	40,472	31,549	6,938	5,208	203,373	35
Ramat Gan	524,401	171,332	72,350	67,448	21,396	16,965	873,891	10
Ramat HaSharon	177,058	61,782	27,959	25,352	7,916	5,646	305,714	27
Ramat Yishay	20,118	16,646	5,651	5,069	937	1,535	49,956	71
Ramla	194,258	88,397	53,484	52,702	8,609	5,156	402,607	22
Rehovot	436,166	172,310	105,059	87,667	17,979	13,135	832,316	11
Rekhasim	16,485	8,692	7,003	8,937	1,145	929	43,190	76
Rishon LeZiyyon	639,306	347,743	143,103	126,393	33,028	25,868	1,315,441	5
Rosh HaAyin	151,745	93,560	37,611	32,416	8,370	5,844	329,546	26
Sakhnin	51,851	42,822	21,171	21,300	3,391	2,706	143,241	46
Sederot	76,728	35,511	22,864	21,980	3,875	2,039	162,997	42
Segev-Shalom	23,569	7,326	5,598	5,410	1,831	1,762	45,497	74
Shefar'am	72,984	50,527	33,381	33,261	5,825	4,083	200,061	37
Shelomi	19,965	8,962	1,856	2,033	733	1,736	35,285	80
Shoham	52,889	34,614	7,308	6,826	3,356	1,922	106,915	54
Tel Aviv - Yafo	1,793,342	576,467	315,197	266,852	80,601	61,090	3,093,549	2
Tel Mond	30,136	21,426	6,580	5,974	2,128	2,079	68,323	63
Tur'an	25,774	21,773	5,263	4,351	1,593	2,631	61,386	66
Umm al-Fahm	73,924	69,753	36,700	30,947	7,156	5,342	223,822	33
Yanuh-Jat	9,520	7,865	5,257	5,075	809	961	29,486	88
Yeroham	19,029	14,424	6,895	5,910	1,520	1,452	49,230	72
Yirka	45,320	18,904	20,722	20,730	2,305	2,344	110,325	53
Yoqne'am Illit	89,627	35,205	14,294	12,102	3,154	3,724	158,107	44

Supplementary Table 4. Emission Factors for Private Transport by Fuel Type, Engine Size, and Specifications (tCO_2eq/km)

	Fuel Type	**Engine Size** (cc)	**Euro Standard**	**Direct EF** (tCO_2eq/km)	**WTT EF** (tCO_2eq/km)
Passenger Cars					
	Petrol	<1400	0	0.00018950	0.00002738
	Petrol	<1400	1	0.00018255	0.00002738
	Petrol	<1400	2	0.00017109	0.00002738
	Petrol	<1400	3	0.00016197	0.00002738
	Petrol	<1400	4	0.00015498	0.00002738
	Petrol	<1400	5	0.00014921	0.00002738
	Petrol	<1400	6	0.00014321	0.00002738
	Petrol	1400-2000	0	0.00031161	0.00003049
	Petrol	1400-2000	1	0.00030027	0.00003049
	Petrol	1400-2000	2	0.00028190	0.00003049
	Petrol	1400-2000	3	0.00026617	0.00003049
	Petrol	1400-2000	4	0.00025804	0.00003049
	Petrol	1400-2000	5	0.00024918	0.00003049
	Petrol	1400-2000	6	0.00024145	0.00003049
	Petrol	>2000	0	0.00031161	0.00003767
	Petrol	>2000	1	0.00030027	0.00003767
	Petrol	>2000	2	0.00028190	0.00003767
	Petrol	>2000	3	0.00026617	0.00003767
	Petrol	>2000	4	0.00025804	0.00003767
	Petrol	>2000	5	0.00024918	0.00003767
	Petrol	>2000	6	0.00024145	0.00003767
	Diesel	<1400	0	0.00020553	0.00002486
	Diesel	<1400	1	0.00010207	0.00002486
	Diesel	<1400	2	0.00009441	0.00002486
	Diesel	<1400	3	0.00010228	0.00002486
	Diesel	<1400	4	0.00011891	0.00002486
	Diesel	<1400	5	0.00011333	0.00002486
	Diesel	<1400	6	0.00010865	0.00002486
	Diesel	1400-2000	0	0.00017891	0.00002672
	Diesel	1400-2000	1	0.00017382	0.00002672
	Diesel	1400-2000	2	0.00016011	0.00002672
	Diesel	1400-2000	3	0.00015127	0.00002672
	Diesel	1400-2000	4	0.00015327	0.00002672
	Diesel	1400-2000	5	0.00014665	0.00002672
	Diesel	1400-2000	6	0.00014024	0.00002672
	Diesel	>2000	0	0.00024379	0.00003035
	Diesel	>2000	1	0.00023649	0.00003035
	Diesel	>2000	2	0.00021750	0.00003035
	Diesel	>2000	3	0.00019968	0.00003035
	Diesel	>2000	4	0.00020669	0.00003035
	Diesel	>2000	5	0.00019326	0.00003035

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	Fuel Type	Engine Size (cc)	Euro Standard	Direct EF (tCO_2eq/km)	WTT EF (tCO_2eq/km)
	Diesel	>2000	6	0.00018155	0.00003035
	LPG	<1400	0	0.00015891	0.00001860
	LPG	<1400	1	0.00017890	0.00001860
	LPG	<1400	2	0.00017680	0.00001860
	LPG	<1400	3	0.00016618	0.00001860
	LPG	<1400	4	0.00015462	0.00001860
	LPG	<1400	5	0.00014905	0.00001860
	LPG	<1400	6	0.00014188	0.00001860
	LPG	1400-2000	1	0.00017890	0.00001860
	LPG	1400-2000	2	0.00017680	0.00001860
	LPG	1400-2000	3	0.00016618	0.00001860
	LPG	1400-2000	4	0.00015462	0.00001860
	LPG	1400-2000	5	0.00014905	0.00001860
	LPG	1400-2000	6	0.00014188	0.00001860
	LPG	>2000	0	0.00017890	0.00001860
	LPG	>2000	1	0.00017890	0.00001860
	LPG	>2000	2	0.00017680	0.00001860
	LPG	>2000	3	0.00016618	0.00001860
	LPG	>2000	4	0.00015462	0.00001860
	LPG	>2000	5	0.00014905	0.00001860
	LPG	>2000	6	0.00014188	0.00001860
	Hybrid	<1400	3	0.00011358	0.00002395
	Hybrid	<1400	4	0.00010868	0.00002395
	Hybrid	<1400	5	0.00010465	0.00002395
	Hybrid	<1400	6	0.00010044	0.00002395
	Hybrid	1400-2000	3	0.00014015	0.00002395
	Hybrid	1400-2000	4	0.00013001	0.00002395
	Hybrid	1400-2000	5	0.00012488	0.00002395
	Hybrid	1400-2000	6	0.00011872	0.00002395
	Hybrid	>2000	4	0.00018082	0.00002894
	Hybrid	>2000	5	0.00017462	0.00002894
	Hybrid	>2000	6	0.00016922	0.00002894
	Electric	<1400	5	0.00008846[a]	0.00000659[b]
	Electric	<1400	6	0.00008846[a]	0.00000659[b]
Motorcycles					
	Petrol	—	—	0.00010870	0.00001935

[a]For EV, this factor captures the associated emissions from the electricity generation process for the electricity consumed by the vehicle

[b]For EV, this factor captures the upstream emissions from the electricity fuel supply chain for the electricity consumed by the vehicle

Supplementary Table 5. Emission Factors for Public transport by Vehicle Type, Route Type, and Traffic Conditions (tCO_2eq/km)

Vehicle Type	**Route Types**	**Traffic Situation**	**Direct EF** (tCO_2eq/km)	**Upstream EF** (tCO_2eq/km)
Bus	Innercity	Stop & Go	0.00173581	0.00029424
	Intercity-regional	Heavy	0.00093798	0.00029424
Minibus	Innercity	Stop & Go	0.00035617	0.00005844
	Intercity-regional	Heavy	0.00020490	0.00005844

Supplementary Table 6. Emission Factors of Consumables (tCO_2eq/ton)

Material	**LCA EF** (tCO_2eq/ton)	**Source**	**Gross EF** (tCO_2eq/ton)	**Custom net EF** (tCO_2eq/ton)
Paper	0.910	DEFRA, 2023	1.576	$\mid -0.666 \mid$
Cardboard	0.802	DEFRA, 2023	0.559	0.243
Plastic	3.102	DEFRA, 2023	0.339	2.763
Metal	5.255	DEFRA, 2023	0.883	4.372
Glass	1.403	DEFRA, 2023	0.395	1.008
Diapers	2.296	Adapted from Cordella et al., 2015	0.053	2.243

Supplementary Table 7. Emission Factors of Wastewater Treatment Plants (WWTPs) (tCO_2eq/m^3)

WTTP name	**Treatment type**	**Direct EF** (tCO_2eq/m^3)	**Energy-related EF** (tCO_2eq/m^3)
Afula	Enhanced (MBT)	0.000791	0.000560788
Akko	Enhanced (MBT)	0.001474	0.000560788
Al Hamra	Low	0.007925	0.000302479
Arad	Tertiary	0.001808	0.000619494
Arei HaEmeq - Yoqne'am Tiv'on	Intermediate	0.002837	0.000302479
Aruar (Ar'ara-BaNegev)	Tertiary	0.000893	0.000619494
Ashdod	Enhanced (MBT)	0.001954	0.000560788
Ashkelon	Enhanced (MBT)	0.000986	0.000560788
Ayalon (Nesher)	Enhanced (MBT)	0.000928	0.000560788
Be'er Sheva	Tertiary	0.000938	0.000619494
Bet She'an	Enhanced (MBT)	0.001580	0.000560788
Bitanya (Alummot)	Enhanced (MBT)	0.002253	0.000560788

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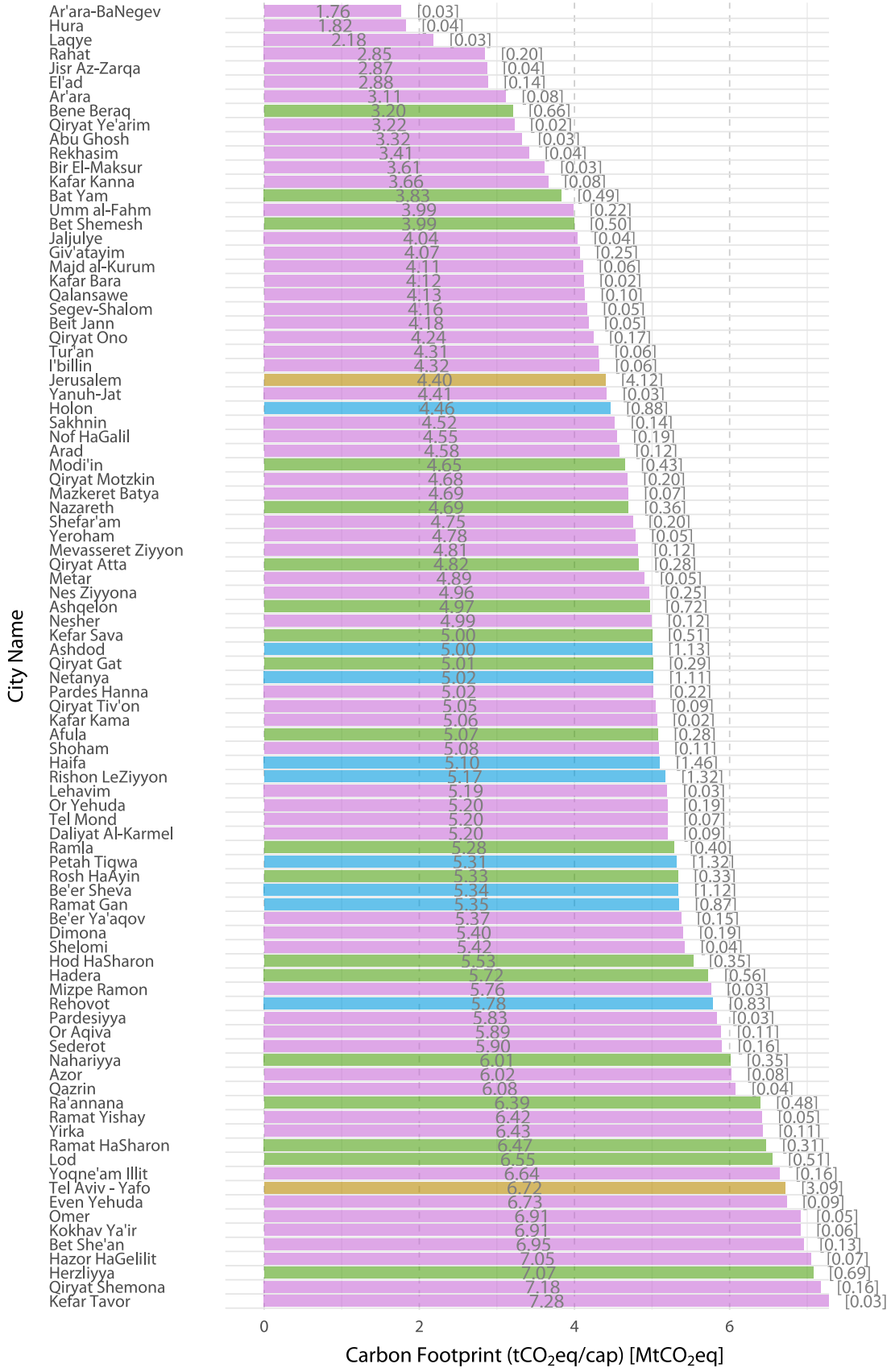
WTTP name	**Treatment type**	**Direct EF** (tCO_2eq/m^3)	**Energy-related EF** (tCO_2eq/m^3)
Darom Sharon Mizrahi	Intermediate	0.003038	0.000302479
Dimona	Tertiary	0.001389	0.000619494
Eilat	Enhanced (MBT)	0.000956	0.000560788
Eyal Tzafon (Tzur Yig'al-Kokhav Ya'ir)	Intermediate	0.003041	0.000302479
Giv'at Brenner	Tertiary	0.001090	0.000619494
Hadera	Enhanced (MBT)	0.000872	0.000560788
HaGoshrim	Intermediate	0.003033	0.000302479
Haifa	Enhanced (MBT)	0.001072	0.000560788
HaSolelim - MBR (Arei HaEmeq)	Enhanced (MBT)	0.001217	0.000560788
HaSolelim (Arei HaEmek)	Intermediate	0.003048	0.000302479
HaSolelim (Weighted)	-	0.002639	0.000366959
Herzliyya	Enhanced (MBT)	0.000641	0.000560788
Horeshim	Enhanced (MBT)	0.001386	0.000560788
Jerusalem (East) - Har Homa	Enhanced (MBT)	0.000992	0.000560788
Jerusalem (East) - Nabi Musa	Intermediate	0.003043	0.000302479
Jerusalem (Weighted)	-	0.001551	0.000543398
Jerusalem (West) - Sorek	Enhanced (MBT)	0.001213	0.000560788
Kafar Manda (Netofa)	Enhanced (MBT)	0.001615	0.000560788
Karmi'el	Tertiary	0.001277	0.000619494
Kefar Sava - Hod HaSharon	Tertiary	0.001075	0.000619494
Kefar Shemaryahu	Low	0.007909	0.000302479
Kefar Tavor	Intermediate	0.003038	0.000302479
Lehavim	Tertiary	0.001004	0.000619494
Livnim	Tertiary	0.001761	0.000619494
Ma'a lot - Ga'ton	Intermediate	0.003070	0.000302479
Ma'yan Zevi - Hof Carmel Dromi	Enhanced (MBT)	0.003649	0.000560788
Menashe 1 - En Shemner (Iron)	Enhanced (MBT)	0.001156	0.000560788
Migdal HaEmeq - Ginnegar (Arei HaEmeq)	Intermediate	0.003235	0.000302479
Mizpe Ramon	Tertiary	0.001047	0.000619494
Nahariyya	Enhanced (MBT)	0.000934	0.000560788
Netanya - Bet Herut	Tertiary	0.001011	0.000619494
Netivot	Enhanced (MBT)	0.001609	0.000560788
Nir Ezyon - Hof Carmel Tzfoni	Enhanced (MBT)	0.000875	0.000560788
Omer	Tertiary	0.000556	0.000619494
Parod	Tertiary	0.001011	0.000619494
Qiryat Gat	Enhanced (MBT)	0.001206	0.000560788
Qiryat Shemona	Intermediate	0.003036	0.000302479

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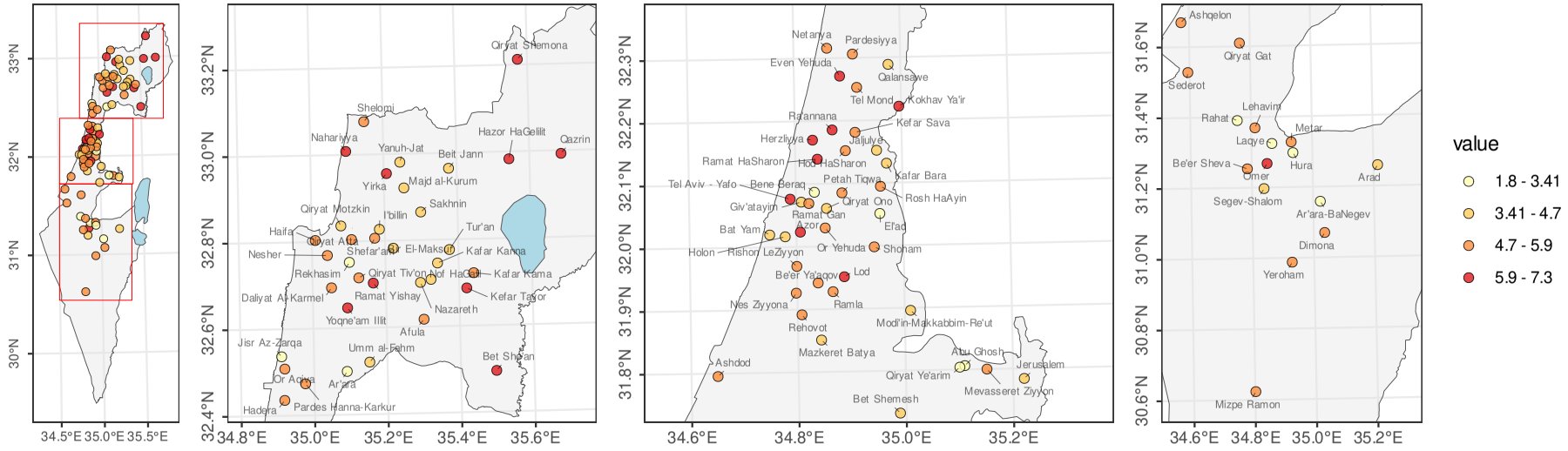
WTTP name	**Treatment type**	**Direct EF** (tCO_2eq/m^3)	**Energy-related EF** (tCO_2eq/m^3)
Ra'annana	Tertiary	0.000716	0.000619494
Rahat	Enhanced (MBT)	0.001918	0.000560788
Ramat HaSharon	Tertiary	0.001018	0.000619494
Ramat Yishay	Intermediate	0.003039	0.000302479
Sde Illan	Tertiary	0.002761	0.000619494
Sderot	Enhanced (MBT)	0.001678	0.000560788
Shafdan	Tertiary	0.001005	0.000619494
Shimshon (Bet Shemesh)	Enhanced (MBT)	0.000696	0.000560788
Shlomi	Intermediate	0.003036	0.000302479
Shoket	Tertiary	0.001027	0.000619494
Tel Mond - Kolhei HaSharon	Enhanced (MBT)	0.001499	0.000560788
Timorim	Enhanced (MBT)	0.002700	0.000560788
Tnuvot	Enhanced (MBT)	0.000762	0.000560788
Yanuh-Jat	Intermediate	0.003047	0.000302479
Yavne	Tertiary	0.001728	0.000619494
Yeroham	Tertiary	0.001194	0.000619494
Zahar (Zefat, Hazor, Rosh Pinna)	Enhanced (MBT)	0.001132	0.000560788
Zefat	Tertiary	0.001494	0.000619494
Zur	Enhanced (MBT)	0.000559	0.000560788

Supplementary Figures

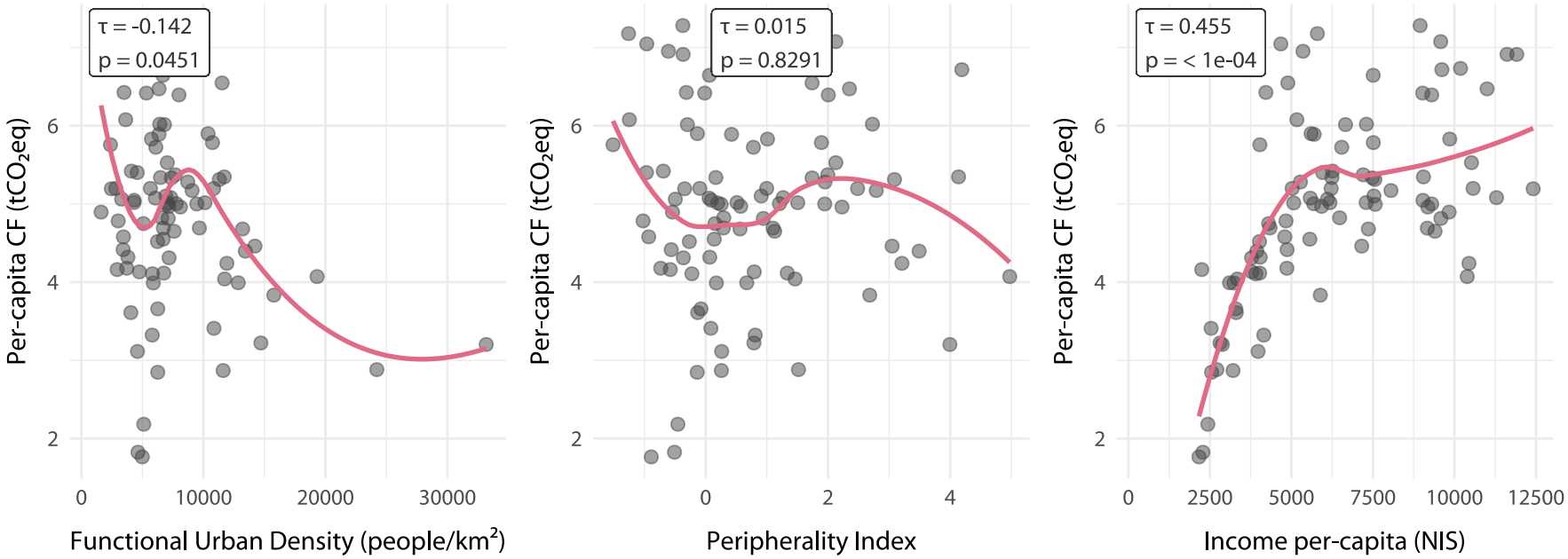
Supplementary Figure 1. Bar Chart: Carbon footprints of analyzed cities: per-capita (tCO_2eq/cap) and Absolute values ($MtCO_2eq$)



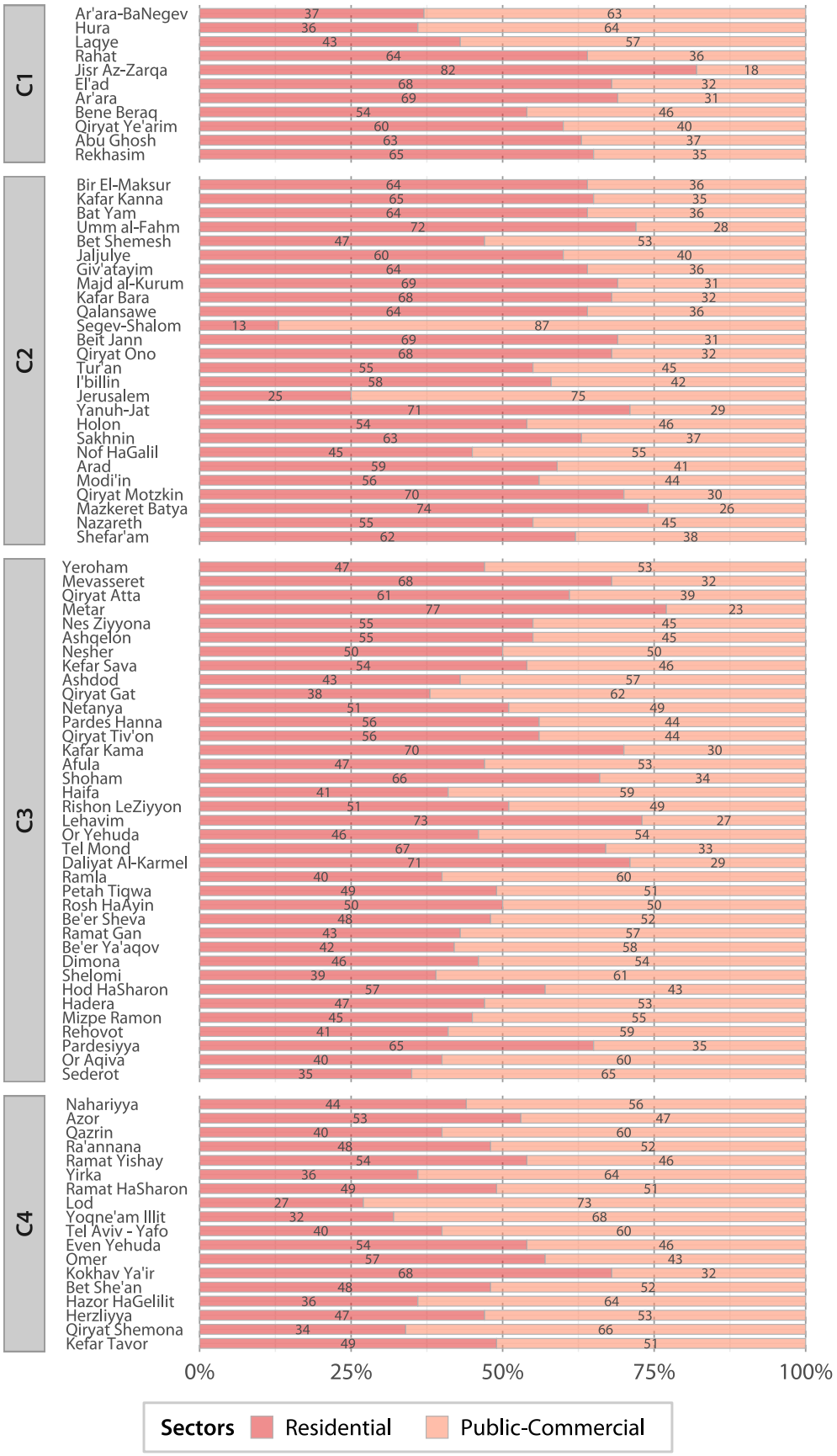
Supplementary Figure 2. Map: Spatial distribution of per-capita carbon footprints of analyzed cities (tCO_2eq/cap)



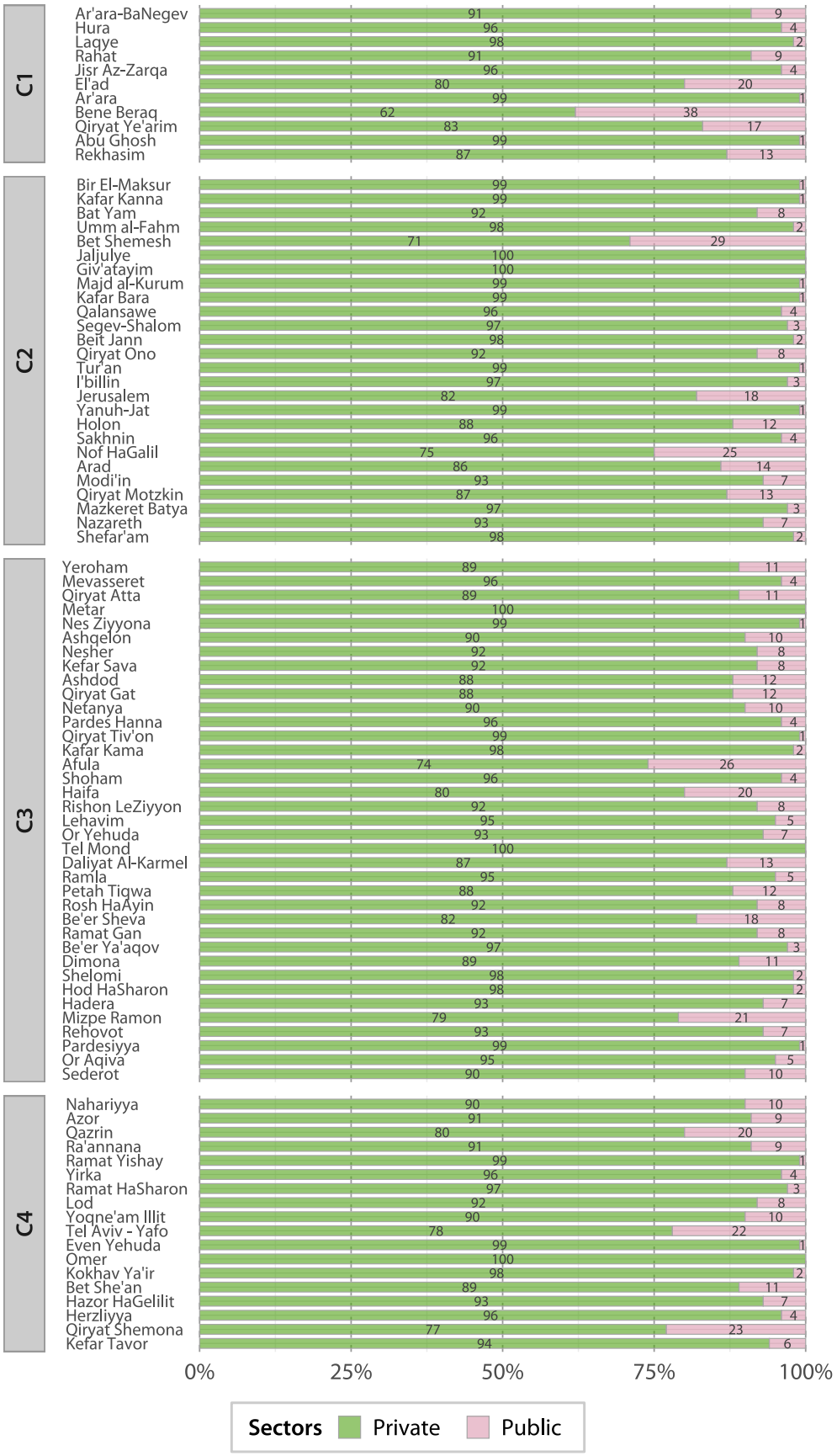
Supplementary Figure 3. Per-capita carbon footprints versus explanatory variables: urban density, peripherality, and income



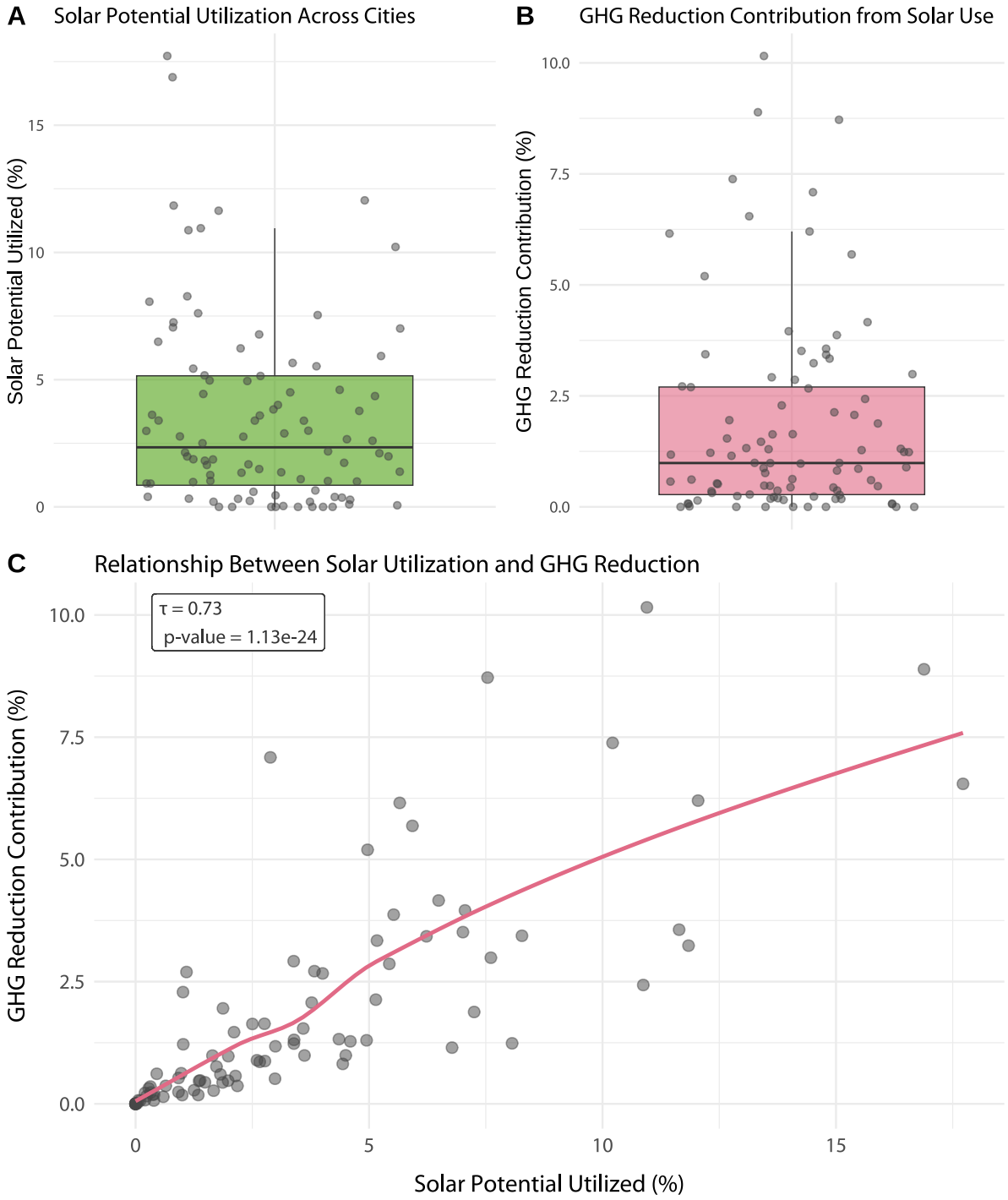
Supplementary Figure 4. Electricity-related emission profiles across cities, by residential and public-commercial sectors (tCO_2eq/cap) [100% Stacked bar chart]



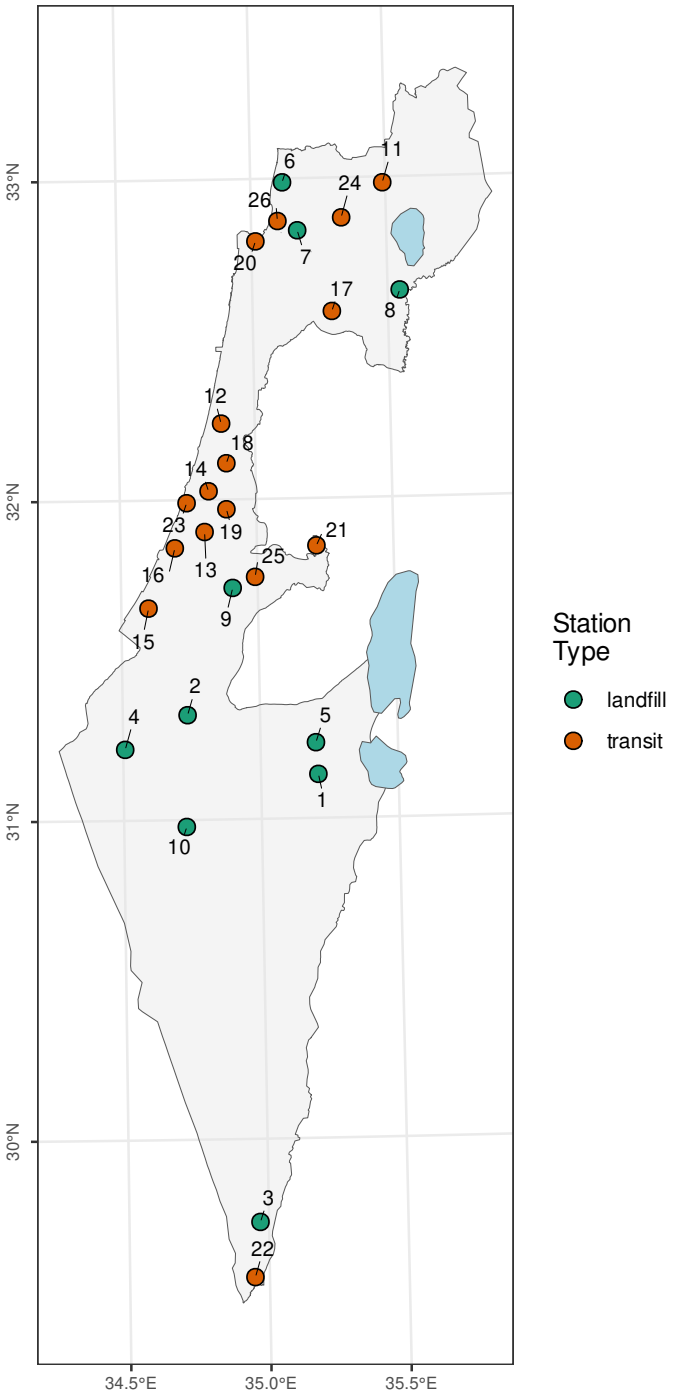
Supplementary Figure 5. Transportation-related emission profiles across cities, by private and public sectors (tCO_2eq/cap) [100% Stacked bar chart]



Supplementary Figure 6. Solar potential utilization and its contribution to GHG reduction across cities

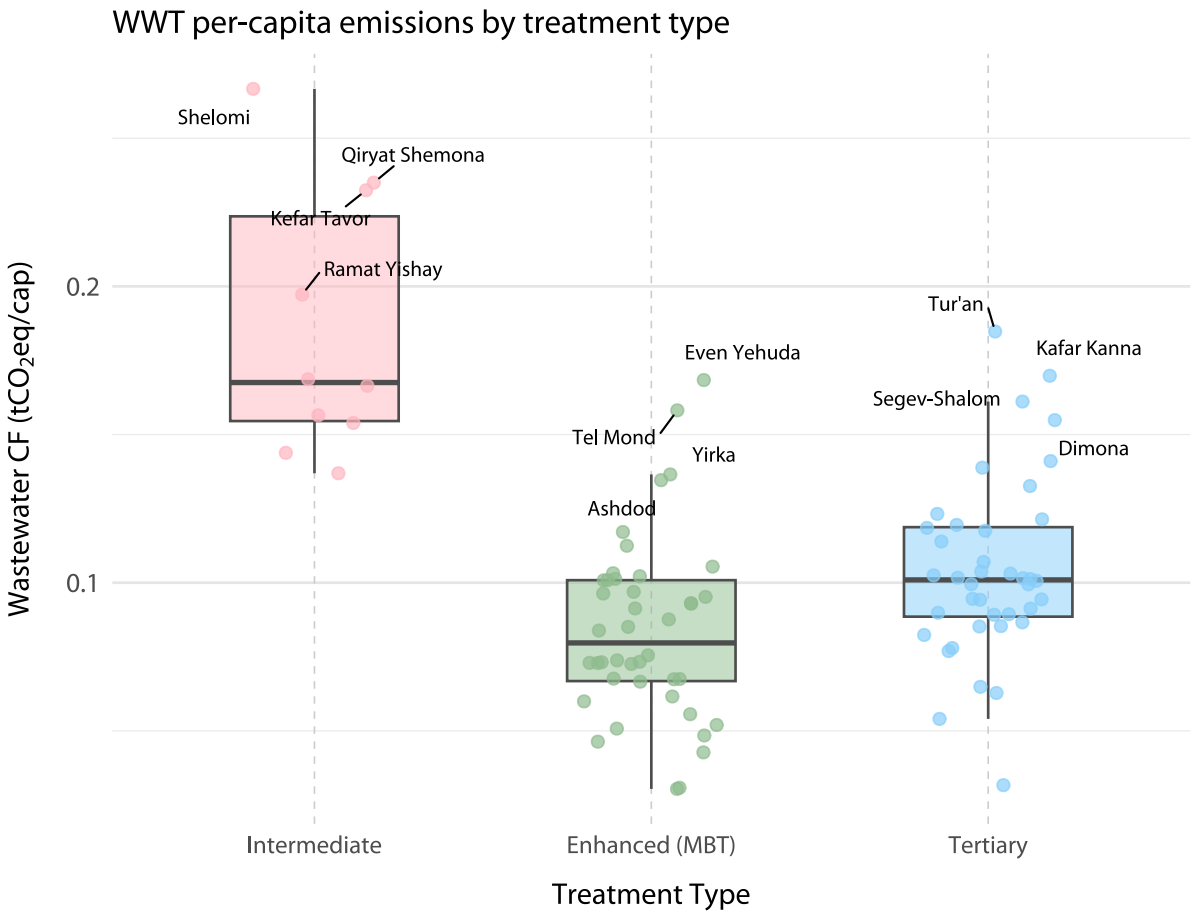


Supplementary Figure 7. Locations of landfills and waste transit stations in Israel



	name_eng	type
1	Ef'a	landfill
2	Duda'im - Ganei Hadas	landfill
3	Nimra	landfill
4	Deia	landfill
5	Zohar	landfill
6	Evron	landfill
7	I'billin	landfill
8	Hagal	landfill
9	Haruvit	landfill
10	Ashalim	landfill
11	Ein Zeitim	transit
12	Mafaat 1965	transit
13	Rehovot	transit
14	Hiriya	transit
15	Ashkelon	transit
16	Ashdod	transit
17	Afula	transit
18	Petah Tikva - Segula	transit
19	Lod	transit
20	Haifa	transit
21	Atarot (Jerusalem)	transit
22	Eilat	transit
23	Rishon LeZion	transit
24	Arraba	transit
25	Beit Shemesh	transit
26	Kiryat Bialik	transit

Supplementary Figure 8. Wastewater treatment emissions of analyzed cities by treatment type (tCO_2eq/cap)



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